

Cascade PID Versus Output-Feedback LQR for the Fixed-Wing Flight Mode of a Hybrid VTOL UAV: Design, Robustness, and a Deterministic Hardware-in-the-Loop Comparison

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Source code and simulation data are available at <https://github.com/Kauemartinsxd/controle-pid-drone-hibrido>.

ABSTRACT This article compares two flight control laws — a classical cascade PID and an output-feedback linear-quadratic regulator (LQRy) — for the fixed-wing flight mode of a small hybrid vertical take-off and landing (VTOL) UAV, following the complete path from design to embedded validation. Both controllers are designed on a model linearized about the cruise trim of a nonlinear model identified from flight-test data, whose open-loop dynamics include an unstable spiral mode, and are evaluated against a common set of literature-grounded closed-loop requirements. In software-in-the-loop (SIL) simulation on the nonlinear model, the two laws are exercised under identical references on a 300-second combined mission; the gain-scheduled LQRy tracks faster and with smaller RMS errors but violates the heading-overshoot requirement, while the requirements-retuned PID meets every requirement at lower bandwidth. A robustness study then reveals complementary weaknesses: under a severe lateral-directional model mismatch both compared laws lose the mission — the aggressive LQRy diverging outright, while an earlier, softer fixed-gain LQRy iteration completes it with near-nominal error, showing that this robustness is set by the tuning rather than by the LQ structure — whereas under a longitudinal-only mismatch the LQRy stays nominal and the PID's clamped altitude loop drifts; under wind and gusts the LQRy concedes altitude where the PID stays firm; both tolerate a plausible +10% identification uncertainty. Finally, both controllers are deployed on an STM32 microcontroller through automatic code generation and compared on a lockstep serial hardware-in-the-loop (HIL) bench that is shown to be bit-deterministic, with, for the synchronized PID deployment, a SIL–HIL residual at the quantization floor of the bus, and a sensitivity study ranking the implementation choices that actually affect embedded agreement. The complete toolchain and data are publicly available.

INDEX TERMS Cascade control, code generation, fixed-wing UAV, hardware-in-the-loop simulation, hybrid VTOL aircraft, LQR control, PID control, robustness, STM32.

I. INTRODUCTION

HYBRID vertical take-off and landing (VTOL) aircraft combine the hovering capability of multirotors with the range and efficiency of fixed-wing flight, and have consequently become one of the fastest-growing classes of unmanned aerial vehicles (UAVs) [1], [2]. The price of this versatility is a control problem that changes structure between

flight regimes, and the literature increasingly treats each regime as a control problem in its own right [3]. This article focuses on the fixed-wing flight mode of a hybrid VTOL platform — the mode in which the vehicle spends most of its mission — and on the complete path from controller design to embedded validation; hover and transition are outside its scope.

Autopilots for small UAVs remain dominated by nested proportional–integral–derivative (PID) loops, a legacy of their simplicity and field-proven robustness [4], and this architecture underpins the open-source flight stacks in widespread use today [5], [6]. Linear–quadratic (LQ) methods are the natural model-based alternative, and comparisons between the two families have accumulated since the seminal indoor quadrotor study of Bouabdallah et al. [7]: on quadrotor platforms [8], [9], on a commercial mini-drone with experimental PID/LQR/MPC benchmarks [10], and, less frequently, on fixed-wing longitudinal dynamics [11]. Three gaps recur in this literature. First, most comparisons target rotary-wing platforms; the fixed-wing mode of a hybrid vehicle is rarely the object of study. Second, the comparisons are usually confined to simulation of isolated channels, rather than a full mission exercising the longitudinal and lateral–directional loops together on the embedded implementation. Third, robustness — arguably where the two design philosophies should differ most — is seldom part of the protocol, even though LQ designs carry classical margin guarantees [12] that do not survive output feedback in general [13].

Hardware-in-the-loop (HIL) simulation — preceded in practice by software-in-the-loop (SIL) verification on the same models — is the established way to close the gap between simulation and flight: the embedded control system is exercised against a simulated plant, so faults in the embedded implementation surface on the bench rather than in the air [14], [15]. The practice extends from classical avionics benches to recent microcontroller-class autopilot platforms [16], [17], is integral to model-based development workflows in which the flight code is generated automatically from the verified model [18]–[20], and has been applied to hybrid VTOL vehicles in the Brazilian research context [21]. Within our research group, this lineage runs through the Pegasus autopilot HIL architecture [22], X-Plane-based HIL of small-UAV attitude control under atmospheric disturbances [23], and the bench-validated fixed-wing autopilot of Santos [24], [25], whose cascade methodology this work adopts. The aircraft models in this line of work are identified from flight-test data [26]–[28].

The present work extends that lineage in three directions: the plant is the fixed-wing mode of a hybrid VTOL aircraft rather than a conventional fixed-wing one; the embedded controller is not hand-coded but automatically generated from the same Simulink model that was verified in simulation, so that the tested artifact is the production code itself; and two different control laws — a cascade PID and an output-feedback LQR (LQRy) — are designed against a common, literature-grounded requirements set, stressed under model mismatch and wind, and deployed on the same bench under identical conditions.

The main contributions of this work are:

- the design of a cascade PID architecture for the fixed-wing flight mode of a hybrid VTOL aircraft, evaluated against a gain-scheduled output-feedback LQR (LQRy) designed for the same platform, both judged against a

common set of literature-grounded closed-loop requirements;

- a robustness study showing that a severe, sign-changing lateral–directional model mismatch defeats both compared laws — the aggressively tuned LQRy diverging outright, while an earlier, softer fixed-gain iteration of the same structure survives it, isolating tuning (not structure) as what buys this robustness — and exposing complementary weaknesses in the remaining scenarios: under a longitudinal-only mismatch the LQRy stays nominal while the PID’s clamped altitude loop drifts, and under wind and gusts the LQRy concedes altitude where the PID stays firm;
- a low-cost HIL architecture based on automatic code generation and a lockstep serial protocol, whose SIL–HIL residual is shown to reach the quantization floor of the communication bus for the synchronized PID deployment;
- the characterization of the lockstep HIL loop as a deterministic system — demonstrated by bit-exact reproduction of full 300-second flight cases across two microcontroller models — and its use to quantify how much each implementation choice affects SIL–HIL agreement; and
- a controlled, like-for-like comparison of the two controllers on the same rig, same references, and same metrics, with the known asymmetries of the two design harnesses disclosed explicitly.

The remainder of this article is organized as follows. Section II presents the nonlinear model, the trim condition, and the linearization. Section III details the closed-loop requirements and both controller designs. Section IV reports the SIL simulations and the controller comparison. Section V presents the robustness study. Section VI describes the HIL development, the determinism and sensitivity analyses, and the embedded results. Section VII concludes the article.

II. NONLINEAR MODEL, TRIM, AND LINEARIZATION

A. NONLINEAR MODEL

The aircraft is a small hybrid VTOL platform with a conventional fixed-wing configuration for cruise. Its physical parameters are $m = 2.2$ kg, wing area $S = 0.27$ m², span $b = 1.2$ m, mean aerodynamic chord $\bar{c} = 0.226$ m (aspect ratio $AR = 5.33$, Oswald factor $e = 0.8$), and inertia tensor (kg m²)

$$\mathbf{I} = \begin{bmatrix} 0.1441 & 0 & -0.00167 \\ 0 & 0.1155 & 0 \\ -0.00167 & 0 & 0.2572 \end{bmatrix}. \quad (1)$$

The complete six-degree-of-freedom nonlinear model of this aircraft [29], [30] follows the standard rigid-body formulation [31] and is implemented in Simulink with a state vector of 14 states (body velocities, angular rates, Euler angles, and navigation states). The aerodynamic model uses the usual

coefficient build-up,

$$\begin{aligned} C_L &= C_{L_0} + C_{L_\alpha} \alpha + C_{L_q} \frac{q\bar{c}}{2V_0}, & C_D &= C_{D_0} + \frac{C_L^2}{\pi e A R}, \\ C_m &= C_{m_0} + C_{m_\alpha} \alpha + C_{m_{\delta_e}} \delta_e + C_{m_q} \frac{q\bar{c}}{2V_0}, \end{aligned} \quad (2)$$

with analogous expressions for the lateral-directional coefficients C_Y , C_l , and C_n in β , p , r , δ_a , and δ_r . The stability and control derivatives were obtained by system identification from flight-test data of the actual aircraft [28] (longitudinal Model 4 and lateral-directional Model 2 of that identification campaign), following the time-domain output-error practice that is standard for flight-vehicle identification [32] and for small UAVs in particular [33]. Thrust is modeled as $T = T_{max} \delta_t (\rho/\rho_0)^{0.8}$ with $T_{max} = 14$ N.

In the fixed-wing flight mode considered here the hover rotors are inactive, so the plant input vector comprises the four aerodynamic commands and the three components of the wind velocity, the latter expressed in the local NED frame:

$$U = \underbrace{[\delta_t \ \delta_e \ \delta_a \ \delta_r]}_{\text{commands}} \mid \underbrace{[w_N \ w_E \ w_D]}_{\text{wind (NED)}}]^T. \quad (3)$$

The wind components are rotated into body axes and subtracted from the inertial velocities before the aerodynamic triad is formed, so that airspeed, angle of attack, and sideslip are computed from the air-relative velocity, $V_T = \|[u \ v \ w]^T - L_{bn}[w_N \ w_E \ w_D]^T\|$, in the usual way [31]. Wind therefore enters the model as a physical disturbance input to the aerodynamics rather than as an additive signal on the outputs; it is identically zero at trim and in the nominal simulations, and is exercised as a disturbance in the robustness study of Section V.

Actuator dynamics are included explicitly, since the controllers are ultimately deployed on hardware: each control surface is limited to $\pm 25^\circ$ in position and $150^\circ/\text{s}$ in rate, followed by a first-order lag of $\tau_{act} = 0.05$ s (20 rad/s bandwidth); the throttle channel is limited to $[0, 1]$ with a rate limit of 1 s^{-1} and a spool-up lag of $\tau_{eng} = 0.30$ s. Without these, a reference step produces an unrealizable impulsive surface deflection ($\sim 400^\circ/\text{s}$ in a single step) that no servo — and no HIL target — could reproduce.

B. TRIM CONDITION

The aircraft is trimmed numerically — a Newton solution of $\dot{x} = f(x, \delta) = 0$ under the flight-condition constraints, the standard formulation of the general aircraft trim problem [31], [34] — in steady, wings-level horizontal flight at airspeed $V_T^e = 12.0$ m/s, altitude $h^e = 600$ m, and flight-path angle $\gamma^e = 0$. The resulting equilibrium is stated explicitly below, as it anchors every simulation and every embedded test in this article.

State vector at trim (body axes; unlisted states are zero):

$$\begin{aligned} u^e &= 11.621 \text{ m/s}, & w^e &= 2.992 \text{ m/s}, \\ \theta^e &= 0.25196 \text{ rad} = 14.44^\circ, & h^e &= 600 \text{ m}, \end{aligned} \quad (4)$$

so that $V_T^e = \sqrt{(u^e)^2 + (w^e)^2} = 12.0$ m/s and, since $\alpha^e = \arctan(w^e/u^e) = 14.44^\circ = \theta^e$, the flight-path angle is indeed $\gamma^e = \theta^e - \alpha^e = 0$.

Output vector at trim:

$$V_T^e = 12.0 \text{ m/s}, \quad \alpha^e = 14.44^\circ, \quad \beta^e = 0, \quad \gamma^e = 0, \quad (5)$$

with $p^e = q^e = r^e = 0$, $\phi^e = \psi^e = 0$, $\theta^e = 14.44^\circ$, and $h^e = 600$ m.

Command vector at trim (control surfaces and throttle):

$$\delta^e = \begin{bmatrix} \delta_t^e \\ \delta_e^e \\ \delta_a^e \\ \delta_r^e \end{bmatrix} = \begin{bmatrix} 0.2836 \\ 0.1330 \text{ rad } (7.62^\circ) \\ 0 \\ 0 \end{bmatrix}, \quad (6)$$

with the hover-rotor commands identically zero in this flight mode. Throughout the article, the controllers act on increments around this trim; the trim commands of (6) are added at the controller output (see the remark in Section IV-A).

C. LINEARIZATION

The nonlinear model is linearized about the trim condition (4)–(6), decoupling into the standard longitudinal and lateral-directional subsystems used for controller design in Section III. With longitudinal states $x_{lon} = [u \ \alpha \ q \ \theta \ h]^T$ and inputs $[\delta_t \ \delta_e]^T$:

$$A_{lon} = \begin{bmatrix} 0.124 & 13.29 & -2.643 & -9.497 & 0 \\ -0.135 & -2.989 & 0.751 & 0 & 0 \\ 0 & -38.75 & -4.108 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -12 & 0 & 12 & 0 \end{bmatrix}, \quad B_{lon} = \begin{bmatrix} 6.071 & 0 \\ -0.126 & 0 \\ 0 & 45.98 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (7)$$

and with lateral-directional states $x_{lat} = [\beta \ p \ r \ \phi \ \psi]^T$ and inputs $[\delta_a \ \delta_r]^T$:

$$A_{lat} = \begin{bmatrix} -0.478 & 0.220 & -0.967 & 0.791 & 0 \\ -51.67 & -12.43 & 9.362 & 0 & 0 \\ 15.74 & -0.648 & -2.355 & 0 & 0 \\ 0 & 1 & 0.257 & 0 & 0 \\ 0 & 0 & 1.033 & 0 & 0 \end{bmatrix}, \quad B_{lat} = \begin{bmatrix} 0 & 0.040 \\ -132.9 & -12.69 \\ 3.282 & 19.09 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (8)$$

(entries below 10^{-3} shown as zero).

D. OPEN-LOOP STABILITY

Table 1 lists the eigenvalues of A_{lon} and A_{lat} and the corresponding classical modes [35], [36]. The longitudinal dynamics are stable, with a well-damped short period and a lightly damped phugoid ($\zeta = 0.085$). The lateral-directional dynamics, however, contain an unstable spiral mode at $s = +0.233 \text{ s}^{-1}$ — an amplitude doubling time of only $t_2 = \ln 2/0.233 \approx 3.0$ s — so the aircraft is open-loop unstable and requires closed-loop stabilization for any practical operation.

This analysis is corroborated by an open-loop simulation of the nonlinear model with the commands frozen at the trim values (6) (Fig. 1). Initialized exactly at the trim state (4), the aircraft remains at equilibrium for the full 300 s (deviations below 10^{-5}), confirming the trim point. Initialized with a small attitude perturbation ($\Delta\theta = 1^\circ$, $\Delta\phi = 5^\circ$) — an offset representative of a minor upset, whose exact size is

TABLE 1. Open-loop modes at the trim condition ($V_T^e = 12$ m/s, $h^e = 600$ m).

Mode	Eigenvalues [s^{-1}]	ω_n [rad/s]	ζ
Short period	$-3.39 \pm 5.31i$	6.30	0.54
Phugoid	$-0.094 \pm 1.11i$	1.12	0.085
Altitude	-1.7×10^{-4}	—	—
Roll subsidence	-11.33	—	—
Dutch roll	$-2.08 \pm 4.59i$	5.04	0.41
Spiral (unstable)	+0.233	—	—
Heading	0	—	—

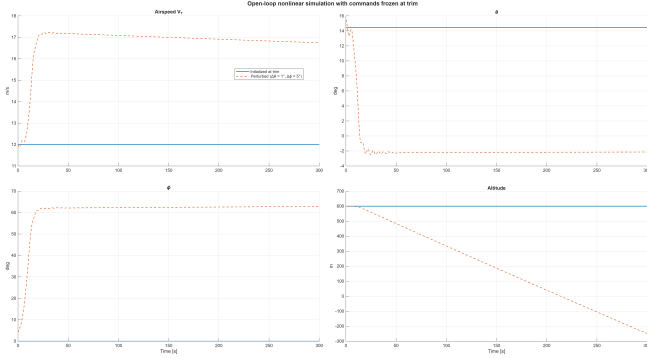


FIGURE 1. Open-loop nonlinear simulation with commands frozen at trim. Solid: initialized exactly at trim (remains at equilibrium). Dashed: small initial perturbation — the unstable spiral mode diverges.

immaterial since the divergence is exponential — the unstable spiral mode takes over: the bank angle diverges to $\approx 63^\circ$; the aircraft accelerates and enters a steadily descending spiral — the behavior predicted by Table 1. No other disturbance is applied; the perturbation acts only through the initial condition.

III. CONTROLLER DESIGN

A. CLOSED-LOOP REQUIREMENTS

Both control laws are judged against the closed-loop requirements of Table 2, adopted from the design practice of our research group [37] and anchored, row by row, in the flight-control literature. The stability-margin rows descend from the military flight-control-system specification MIL-F-9490D, which requires a gain margin of at least 6 dB and a phase margin of at least 45° for the augmented airplane [38] — limits carried into present-day industry practice by its successor SAE AS94900 [39]; the 10 dB adopted here is deliberately conservative. The damping row is stricter than the flying-qualities levels of MIL-F-8785C and MIL-STD-1797A, whose Level-1 bands admit short-period and Dutch-roll damping well below 0.75 [40]–[42]; the $\zeta > 0.75$ target reflects the classical well-damped second-order response ($\zeta \approx 0.7$ yields $\approx 5\%$ overshoot [43]). The time-domain rows (settling time, overshoot, steady-state error) are outer-loop performance targets in the sense of classical autopilot design texts [44], [45], sized for this platform’s mission profile. Since these criteria were written for piloted aircraft, their transfer to a small UAV is itself a design decision, supported

TABLE 2. Closed-loop design requirements and their scope.

Requirement	Value	Scope
Settling time	< 60 s	guidance channels (V_T, ψ, h)
Overshoot	$< 10\%$	guidance channels (V_T, ψ, h)
Steady-state error	2–5%	guidance channels (V_T, ψ, h)
Damping ratio	$\zeta > 0.75$	augmented inner modes
Gain margin	> 10 dB	each loop, at design
Phase margin	$> 45^\circ$	each loop, at design

by the literature on scaled flying-qualities requirements for unmanned aircraft [46].

Each row carries an explicit scope. The time-domain rows apply to the guidance-level channels (V_T, ψ, h) on shaped (prefiltered) mission references — they are verified against both controllers in Section IV-E. The damping row applies to the augmented inner modes at design time; the margin rows are enforced per loop during loop shaping (the phase-margin targets of Table 3, 55 – 65° , exceed the requirement). One honest caveat closes the picture: the augmented Dutch-roll damping achieved by the PID design (0.707, Section III-B) satisfies the MIL-F-8785C Level-1 band with wide margin but sits marginally below the stricter internal target of 0.75 — reported as designed rather than adjusted to fit. On the short-period side, the pitch-rate damper gain of the tuning of record ($K_q = 0.10$, Section III-B) was raised precisely to lift the short-period damping.

B. CASCADE PID ARCHITECTURE

The PID controller follows a classical cascade architecture — successive loop closure in the sense of [47] — designed loop by loop on the linearized models of Section II and then verified on the full nonlinear model. The design methodology — loop ordering, damper-first stability augmentation, and the coordinated-turn heading law — follows [24], developed there for a conventional general-aviation aircraft and applied here to the hybrid VTOL platform in its fixed-wing flight mode.

1) Stability Augmentation

Rate dampers are closed first, tuned by root locus on the linear subsystems. The pitch-rate damper $K_q q$ raises the short-period damping ($K_q = 0.10$ in the tuning of record, a deliberately raised damping target); the yaw damper feeds r through a washout $s/(s+1)$ to the rudder, raising Dutch-roll damping to $\zeta_{DR} \approx 0.707$ without opposing steady turns. The washout yaw damper is one of the oldest devices of automatic flight control: introduced with the first swept-wing jets — which could not be flown unaugmented because of their Dutch roll — in the late 1940s [48], it received its definitive analytical treatment in the classical stability-augmentation literature [44], [49], which the present implementation follows. The roll subsidence mode already meets Level 1 handling qualities with margin ($\lambda = -11.3 s^{-1}$, Table 1), so no roll-rate damper is used ($K_p = 0$).

TABLE 3. PID loops: design targets and resulting gains (tuning of record).

Loop	Map	K_P	K_I	Original target
C_θ	$\theta^{ref} \rightarrow \delta_e$	0.917	0.857	$\omega_c=2.0$, PM=65°
C_{vel}	$V_T^{ref} \rightarrow \delta_t$	0.32	0.096	$\omega_c=0.8$
C_{alt}	$h^{ref} \rightarrow \theta^{ref}$	0.050	0.0043	$\omega_c=0.35$, PM=55°
C_ϕ	$\phi^{ref} \rightarrow \delta_a$	-0.45	-0.036	$\omega_c=2.0$, PM=65°
$K_{heading}$	$\psi^{ref} \rightarrow \phi^{ref}$	0.1975	—	$\tau_\psi=6$ s
K_q	$q \rightarrow \delta_e$	0.10	—	ζ_{SP} raised
K_r	$r \rightarrow \delta_r$ (washout)	0.1481	—	$\zeta_{DR}=0.707$

2) Longitudinal Loops

- Pitch-attitude inner loop (C_θ): PI on $\theta^{ref} - \theta$ to the elevator, closed over the K_q damper;
- Airspeed hold (C_{vel}): PI on $V_T^{ref} - V_T$ to the throttle;
- Altitude outer loop (C_{alt}): PI on $h^{ref} - h$ to the pitch-attitude reference, with the command clamped to $\pm 10^\circ$ and clamping-type anti-windup. The clamp is an angle-of-attack protection: with θ^{ref} limited to $+10^\circ$ about the trim, the peak α during an altitude capture stays near 16° (the plant model has no stall representation, so the limit is imposed by design).

Depending on the flight mode, the pilot-level selector engages either the altitude cascade (AltitudeHold) or direct pitch tracking (ThetaHold), in which case the altitude loop is open.

3) Lateral-Directional Loops

- Bank inner loop (C_ϕ): PI on $\phi^{ref} - \phi$ to the aileron (negative gains, since $C_{l_{\delta_a}} < 0$);
- Heading hold ($K_{heading}$): proportional law $\phi^{ref} = K_{heading}(\psi^{ref} - \psi)$ with $K_{heading} = u^e/(g\tau_\psi)$ and $\tau_\psi = 6$ s, the coordinated-turn relation [24]. As with the longitudinal channel, the mode selector engages either heading hold (PsiHold) or direct bank commands (PhiHold).

4) Tuning and Resulting Gains

The compensators are PI ($K_d = 0$): damping is provided by the explicit dampers rather than by derivative action, which also avoids differentiating noisy rate measurements on the embedded target. Each loop was tuned by loop shaping (pidtune) with sequential inner-first closure — every outer loop is designed against the already-closed inner dynamics — targeting the crossover frequencies and margins listed in Table 3, with a bandwidth separation of 5–7 $\times$ between consecutive loops.

The outer altitude and airspeed loops use two-degree-of-freedom setpoint weighting with $b = 0$ — the proportional term acts on the measured output only, with the reference entering through the integral path — and this tuning of record, retuned against the requirements of Table 2, is the configuration used in every campaign of this article, matching the public repository HEAD.

A command prefilter $1/(\tau_{ref}s + 1)$ with $\tau_{ref} = 3$ s is applied to the increment of each reference before the loops,

giving a two-degree-of-freedom structure in the tradition of reference command shaping [50]: the command response is smoothed — the raw step would otherwise demand an unrealizable actuator jump — without altering the feedback design, its margins, or the clamp. This detail matters for the embedded deployment: in the HIL architecture of Section VI the references travel through the link already filtered, so the prefilter stays on the host and only the feedback path is embedded.

With all loops closed, the linear closed-loop system is stable with every pole in the left half-plane: the slowest mode is at -0.127 s $^{-1}$ and the minimum damping ratio among the oscillatory modes is $\zeta = 0.55$. Notably, the open-loop unstable spiral mode ($+0.233$ s $^{-1}$, Table 1) is stabilized at -0.143 s $^{-1}$.

C. OUTPUT-FEEDBACK LQR (LQRY)

The second control law is a linear-quadratic design with output weighting (LQRY): for each loop, the quadratic cost

$$J = \int_0^\infty (y^\top Q y + u^\top R u) dt, \quad y = C_{syn} x, \quad (9)$$

is minimized over the linearized models of Section II, yielding constant gains on the measured states plus integral action on the tracked error. This is the classical linear-quadratic regulator of Kwakernaak and Sivan [51] in the constant-gain output-feedback setting introduced by Levine and Athans [52] — a problem whose difficulty and solution landscape are surveyed in [53], and whose standard numerical solution is the flight-control-motivated iterative algorithm of Moerder and Calise [54]; the formulation and its application to aircraft control follow the treatment of Stevens et al. [31]. The weight-selection rationale follows established LQ design practice [55], starting from Bryson's rule [56]. The design compared in this article is the gain-scheduled LQRY of the companion article by the second author [37], which details its synthesis — including the target-transmission-zero augmentation and the gain scheduling over the flight envelope — and its frequency-domain (Nichols) robustness assessment; it is summarized here.

The architecture deliberately mirrors the PID mode structure, so that the two laws face the same tasks: five independent designs — ThetaHold, AltitudeHold (cascading into θ^{ref}), VelHold, PhiHold (a 2×2 design commanding aileron and rudder), and PsiHold (cascading into ϕ^{ref}) — selected by the same three mode switches used by the PID (Table 4). Each synthesis model augments the relevant linear subsystem with a frequency-selective filter

$$G(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{s} \quad (10)$$

on each tracked output — imposing target transmission zeros that shape the closed loop while providing the integral action — plus a first-order actuator lag at 24 rad/s; the constant output-feedback gains are obtained from the coupled Lyapunov equations of the LQ output-feedback problem [52],

TABLE 4. LQRy loops: regulated output and actuator (weights per envelope condition in the companion article [37]).

Loop	Output	Actuator
ThetaHold	θ	elevator
AltitudeHold	$h \rightarrow \theta^{ref}$	elevator
VelHold	V_T	throttle
PhiHold	ϕ, β	ail.+rud.
PsiHold	$\psi \rightarrow \phi^{ref}$	aileron

TABLE 5. LQRy gains at envelope condition 2 (12 m/s, 600 m), as used in the SIL comparison (4 significant digits). Convention: $u = G_x x + K_I \int e dt$. The K_I column of PhiHold is the bank (ϕ) integral gain.

Loop	State vector x	G_x	K_I
ThetaHold	$[V_T \alpha q \theta \delta_e]$	$[-0.00461 \ 6.477 \ -15.60 \ -483.2 \ -6.795]$	-625
AltitudeHold	$[V_T \alpha q \theta h]$	$[-0.0718 \ 4.474 \ -0.2121 \ -5.064 \ -0.5086]$	-0.2025
VelHold	$[V_T \alpha q \theta]$	$[-78.81 \ -0.9055 \ 0.0589 \ 1.306]$	-50
PhiHold (ail.)	$[\beta pr \phi]$	$[0.00693 \ 0.01013 \ 0.03505 \ 0.1595]$	0.1720
PhiHold (rud.)	$[\beta pr \phi]$	$[-0.01085 \ -0.02236 \ -0.08167 \ -0.3542]$	-0.3595
PsiHold	$[\beta pr \phi \psi]$	$[-5.387 \ -0.1123 \ -0.7948 \ -1.272 \ -6.037]$	-7.773

[54]. The design is synthesized at nine envelope conditions (12/15/18 m/s $\times$ 590/600/610 m); the SIL comparison of Section IV exercises the condition-2 (12 m/s, 600 m) gain set, consistent with the mission's initial trim — mid-mission gain switching is not exercised. The weights per condition are given in the companion article [37].

Table 5 records the condition-2 gains: these are the values used in the SIL comparison of Section IV. Two structural remarks: the pitch design feeds back the elevator (actuator) state itself, its synthesis model carrying the 24 rad/s lag as a fifth state; and the 2×2 lateral design carries a sideslip-integral gain alongside the bank integral (0.00845 on the aileron and 0.02023 on the rudder at condition 2) — small relative to the ϕ integral, but not negligible. The earlier fixed-gain LQRy delivery, deployed on the HIL bench of Section VI, used a different, softer gain set, available in the repository [30].

Unlike the PID harness, the LQRy harness of the gain-scheduled design includes first-order actuator lags (24 rad/s on the surfaces, 0.1 s on the throttle) and a $[0, 1]$ throttle saturation, but no rate limits, no surface position saturations, no reference prefilter, and no θ^{ref} clamp — references enter the loops as raw steps. These differences between the two design harnesses are consolidated and discussed in Section IV-B.

D. ROLL-TO-LONGITUDINAL COUPLING

Both designs treat the longitudinal and lateral-directional channels independently, so the classical roll-to-pitch coupling — banking tilts the lift vector, and the lift deficit, proportional to $1 - \cos \phi$, drops the nose and costs altitude — is handled by feedback alone. Since this is a deliberate design choice, it was quantified rather than assumed benign: both controllers — the PID tuning of record and the gain-scheduled LQRy — were subjected to bank captures of increasing amplitude (ϕ^{ref} steps of 5/15/30° at $t = 20$ s, each rig shaping the reference as its harness prescribes), in ThetaHold (altitude loop open, the worst case) and in AltitudeHold, with all longitudinal metrics taken as deviations from each rig's own $\phi = 0$ baseline (Fig. 2).

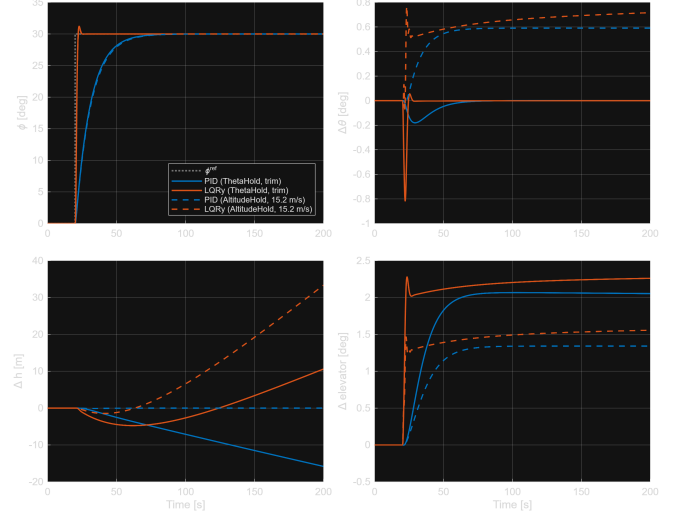


FIGURE 2. Roll-to-longitudinal coupling, $\phi^{ref} = 30^\circ$ step at $t = 20$ s: ϕ (with the reference), $\Delta\theta$, Δh , and $\Delta\text{elevator}$ relative to each configuration's $\phi = 0$ baseline. Solid: ThetaHold at the 12 m/s trim; dashed: AltitudeHold at 15.2 m/s.

tudeHold, with all longitudinal metrics taken as deviations from each rig's own $\phi = 0$ baseline (Fig. 2).

The results confirm the mechanism and bound its size. For the PID in ThetaHold, the transient pitch dip scales with $1 - \cos \phi$ as expected — -0.005° , -0.04° , and -0.18° for 5/15/30° — negligible at the 5° amplitude of the flight campaign. The LQRy dip is 4–5 $\times$ larger ($\approx -0.8^\circ$ at 30°): its multivariable structure does not natively absorb the coupling better than the cascade. With the altitude loop open the altitude drifts by construction — -15.8 m for the PID over 200 s at 30°, while the LQRy, whose high-bandwidth airspeed loop adds energy through the turn, drifts $+10.6$ m upward; closing the altitude loop absorbs the coupling, with θ^{ref} settling higher ($+0.59^\circ$ PID, $+0.71^\circ$ LQRy) and Δh returning to zero after a transient dip of 0.10 m (PID) and 1.45 m (LQRy), at the cost of a small steady elevator increment.

In the steady turn, the elevator increment is larger in the ThetaHold runs at the 12 m/s trim (solid traces, 2.06° PID and 2.26° LQRy) than in the AltitudeHold runs at 15.2 m/s (dashed, 1.34° and 1.55°), which at first seems backwards — why would the altitude-holding configuration need less elevator? The explanation is the operating point, not the mode: sustaining the same 30° bank (load factor $1/\cos 30^\circ = 1.15$) requires an increment of lift that, at 15.2 m/s, is produced with less angle of attack — and less elevator — because the dynamic pressure is $(15.2/12)^2 = 1.6$ times higher. The measured increment ratios (0.65 for the PID, 0.69 for the LQRy) sit within a few percent of the dynamic-pressure prediction $(12/15.2)^2 = 0.62$, the residual owing to the θ^{ref} adjustment and the angle-of-attack dependence of drag.

In answer to the natural question — no, no feedforward decoupling term is used by either law ($K_{ff} = 0$ throughout): with the altitude loop closed the residual coupling is 0.10 m even at a 30° bank (PID), and at the 5° amplitude of the flight campaign the $1 - \cos \phi$ factor is a further 35 $\times$ smaller, so

feedback suffices. A feedforward stage, and its evaluation on both laws at larger bank angles, is left as future work (Section VII).

IV. CLOSED-LOOP SIMULATIONS (SIL)

A. REMARK ON TRIM COMMANDS

All controllers in this work operate on incremental commands about the trim condition of Section II-B: the compensators output $\Delta\delta$, and the trim commands δ^e of (6) are added before the actuator models. Consequently, whenever a simulation reports a command signal, the reader should distinguish the absolute command $\delta = \delta^e + \Delta\delta$ from the increment $\Delta\delta$ produced by the controller. In steady flight at the trim point, the absolute commands settle at δ^e (e.g., the elevator at 7.62°), not at zero.

B. COMPARISON PROTOCOL AND KNOWN ASYMMETRIES

Every head-to-head result in this article follows the same protocol — in the spirit of the experimental controller benchmarks of the recent literature [9], [10]: the same nonlinear plant, the same reference amplitudes and timings, and the same metric windows, with each control law running in its own design harness, exactly as delivered. That last clause matters, because the two harnesses are not identical, and an honest comparison must say so. Three asymmetries are known and uncontrolled:

- 1) Actuator models. The PID harness includes the full actuator chain of Section II (position and rate saturations, first-order servo and engine lags). The LQRy harness includes first-order actuator lags — 24 rad/s on the surfaces versus the 20 rad/s assumed on the PID side, and 0.1 s versus 0.30 s for the throttle — and a $[0, 1]$ throttle saturation, but omits the rate limits and surface position saturations of the PID harness.
- 2) Reference shaping. The PID applies the $1/(3s + 1)$ prefilter of Section III-B to every reference increment; the LQRy receives raw reference steps.
- 3) Protection logic. The $\pm 10^\circ$ θ^{ref} clamp and its anti-windup exist only in the PID cascade.

To bound the effect of these asymmetries on the conclusions, the PID scenarios were repeated in an equalized configuration — actuator bandwidth matched to 24 rad/s, rate limits removed, clamp and prefilter disabled, matching the LQRy harness. Fig. 3 overlays the two PID configurations on the combined scenario: the equalization shifts the guidance captures moderately — airspeed rise time from 9.5 to 5.8 s and settling from 18.0 to 11.4 s, heading settling from 33.0 to 29.3 s, altitude settling from 44.4 to 40.8 s, all with zero overshoot — without changing any comparative conclusion drawn below.

Two further implementation notes complete the disclosure. First, an earlier fixed-gain LQRy build fed back the absolute state instead of the deviation from trim, producing a start-up kick (initial elevator command of 4.11° instead of the trim

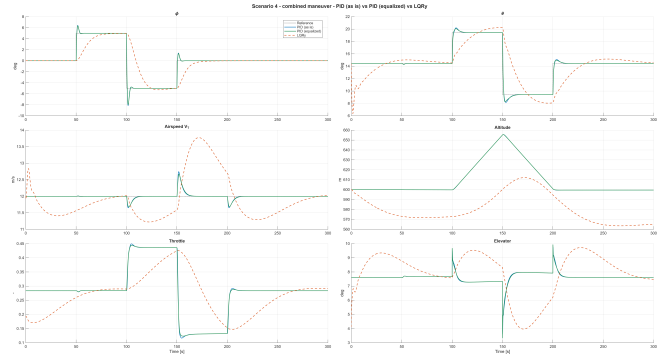


FIGURE 3. Equalization ablation, combined scenario: PID as delivered vs. PID with the harness equalized to the LQRy rig (24 rad/s actuators, no rate limits, no clamp, no prefilter), with the LQRy response for context.

7.62°); the gain-scheduled build compared here feeds back trim deviations by construction, and its runs were verified to start at the trim command. The embedded (incremental-form) implementation of Section VI was immune to this distinction throughout. Second, the LQRy rig initializes the plant at $V_T = 12.0$ m/s while its stored controller trim corresponds to 11.62 m/s, so LQRy runs begin with a small longitudinal offset that the airspeed loop resolves before the first reference event.

C. PID ON THE NONLINEAR MODEL

The cascade PID was evaluated on the nonlinear model under three maneuvers, with the linear closed-loop response overlaid in each case: a longitudinal climb ($h + 10$ m), a lateral-directional heading change ($\psi + 15^\circ$), and an aggressive combined maneuver ($h + 30$ m and $\psi + 40^\circ$ commanded simultaneously, saturating the θ^{ref} clamp for ≈ 10 s). Fig. 4 shows the combined case. Linear and nonlinear responses agree closely even in the aggressive maneuver (maximum deviations of 0.38 m in altitude, 0.52° in pitch, and 0.27° in bank; below 0.21 m/ 0.75° / 0.05° in the single-axis maneuvers), validating both the linearization and the linear-design-nonlinear-verification workflow. This validation was performed with the original loop-shaping tuning; its conclusion — the fidelity of the linearization — is independent of the subsequent requirements-driven retune.

D. LQRy ON THE NONLINEAR MODEL

The LQRy was evaluated directly on the nonlinear model, in its own harness, under the same maneuver set used for the comparison below. Its closed-loop character is consistent across maneuvers: fast, high-bandwidth captures — airspeed and altitude in ≈ 6 s with essentially no overshoot, heading in ≈ 13 s with visible overshoot (Table 6). Unlike the PID case, no linear-overlay validation is presented: the linear design models of Section III-C are the basis of the synthesis, and the nonlinear evaluation here is the verification step of record for the delivered design.

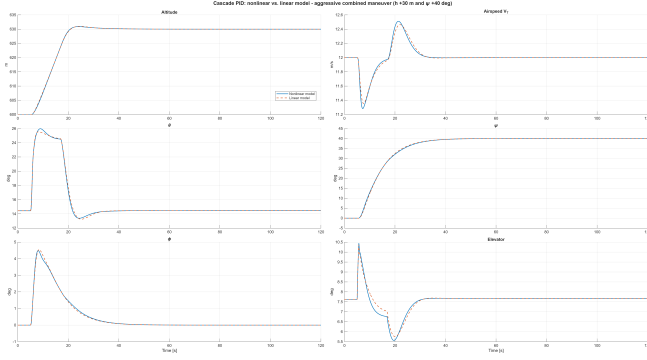


FIGURE 4. PID, aggressive combined maneuver ($h + 30$ m and $\psi + 40^\circ$ simultaneous): nonlinear model vs. linear closed loop.

E. CONTROLLER COMPARISON

The evaluation maneuvers deserve a word of justification. Doublets are the conventional, well-characterized excitation of flight-test practice — originally for identification, where their frequency content is matched to the modes of interest [57], [58] — and step and doublet captures underlie the demonstration requirements of the flying-qualities specifications [40]. For unmanned aircraft, however, the literature argues that mission-based evaluation is the most transferable criterion, since manned handling-qualities metrics do not carry over directly [59]; command-profile tracking of airspeed, altitude, and attitude remains the standard validation practice for small-UAV control laws [60]. The campaign therefore uses both: per-axis guidance scenarios to characterize each channel, and a 300 s combined mission (airspeed step, heading doublet, altitude doublet, overlapping in time) as the mission-level verdict.

The two controllers are compared on the nonlinear model under four guidance-level scenarios (VelHold+PsiHold+AltitudeHold) with identical reference profiles on both rigs (amplitudes injected into each rig's unmodified reference harness):

- 1) Airspeed: step $12 \rightarrow 15.2$ m/s at 5 s;
- 2) Heading: doublet of $\pm 5^\circ$ at 50/100/150 s;
- 3) Altitude: doublet of ± 5 m at 80/160/240 s;
- 4) Combined mission: all three simultaneously, 300 s (flight case 4 of the campaign of Section VI).

Table 6 reports the classical step-response metrics per guidance channel, and Fig. 6 shows the combined mission. The roles observed in the earlier fixed-gain iteration of this comparison are reversed: the gain-scheduled LQRy is now the fast, high-bandwidth law — it captures heading in ≈ 13 s at the cost of 16.4% overshoot, and completes the airspeed and altitude captures in ≈ 6 s with essentially no overshoot — while the requirements-retuned PID is the smooth one, with zero overshoot on every channel and settling times of 18–44 s. Neither tuning dominates: the LQRy buys speed at the cost of a heading-overshoot requirement violation; the PID buys strict compliance at the cost of bandwidth.

Table 7 closes the loop with the requirements of Section III-A, evaluated on the guidance channels of the com-

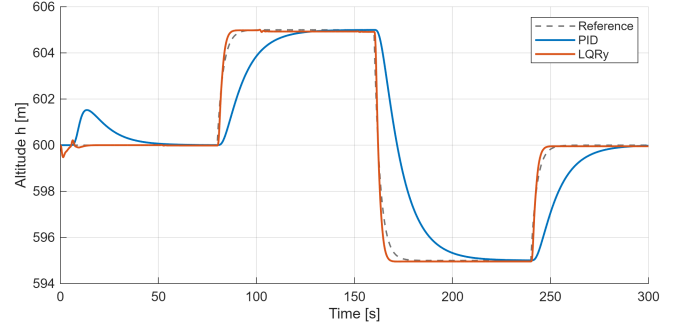


FIGURE 5. Altitude channel of the combined mission (SIL): reference, PID, and LQRy on a single axis.

TABLE 6. Per-axis guidance captures (SIL, identical references), metrics on the +step segment of each channel. OS = overshoot, t_r = 10–90% rise time, t_s = 2% settling time, e_{ss} = steady-state error.

Ch.	OS[%]	PID				LQRy			
		t_r [s]	t_s [s]	e_{ss} [%]		t_r [s]	t_s [s]	e_{ss} [%]	
V_T	0.0	9.48	18.0	0.00		3.04	5.8	0.00	
ψ	0.0	16.14	33.0	0.50		0.83	12.6	0.01	
h	0.0	23.37	44.4	0.06		3.44	5.9	0.19	

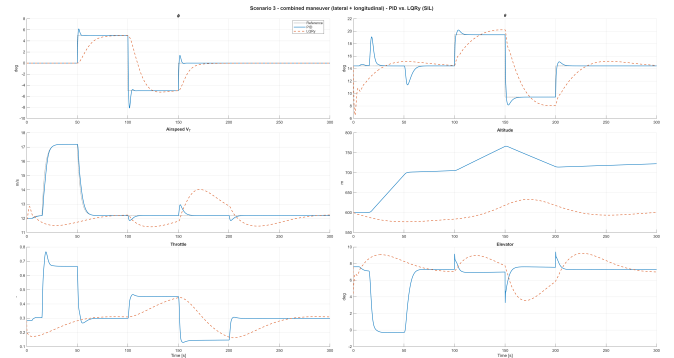


FIGURE 6. Scenario 4 (combined mission): PID vs. LQRy on the nonlinear model, identical references.

TABLE 7. Requirements compliance on the combined mission (SIL), guidance channels, identical references.

Requirement	PID			LQRy		
	V_T	ψ	h	V_T	ψ	h
OS < 10%	0.0	0.0	0.0	0.2	24.5	0.2
t_s < 60 s	18.0	38.1	45.1	5.8	6.9	7.0
$e_{ss} \leq 5\%$	0.00	0.72	0.07	0.00	0.00	1.75

combined mission. The PID meets every row. The LQRy settles far inside the 60 s limit on every channel (5.8–7.0 s) and meets the steady-state row, but violates the overshoot requirement on the heading channel (24.5% against the 10% line).

The embedded (HIL) campaign of Section VI was executed with the earlier fixed-gain LQRy build (Section III-C), so its LQRy-side numbers characterize that deployment rather than the gain-scheduled law compared here; the PID-side SIL–HIL fidelity results are unaffected.

V. ROBUSTNESS STUDY

The comparison of Section IV-E holds for the nominal plant — the very model both controllers were designed on. This section asks how much of that picture survives when the plant is no longer the design model. The protocol is deliberately strict: only the plant changes. Both controllers keep the gains and trim commands of the nominal design, and every run repeats the combined mission of Section IV-E (V_T 12 $\rightarrow$ 15.2 m/s at 5 s, heading doublet $\pm 5^\circ$ at 50/100/150 s, altitude doublet ± 5 m at 80/160/240 s, 300 s total) — the same combined mission, and the same per-axis reference set, used throughout Section IV. Four scenarios are considered, spanning plausible-to-severe model uncertainty and an exogenous disturbance: (A) a severe model mismatch, replacing the complete aerodynamic set by an independent identification of the same airframe; (B) the longitudinal half of that mismatch only, to isolate the origin of the failure observed in (A); (C) a plausible identification uncertainty, multiplying all 23 stability and control derivatives by 1.10; and (D) wind and gusts applied through the native disturbance inputs of (3). Table 8 summarizes the tracking metrics; the degradation columns normalize each scenario by the same controller's nominal performance, so that the two laws are compared against themselves rather than against each other.

A. SEVERE MODEL MISMATCH

The perturbed plant replaces the nominal, flight-test-identified aerodynamic set [28] by the complete alternative set originally distributed with the nonlinear model implementation [29], [30], obtained through an independent identification of the platform. The two sets differ substantially — the pitch-damping derivative C_{m_q} by a factor of 4, the elevator effectiveness $C_{m_{\delta_e}}$ by 3.3, and the lateral-directional set even changes the sign of C_{y_β} and C_{y_p} — so this scenario probes well beyond parametric uncertainty: it is a structurally different aerodynamic description.

Under this mismatch both laws lose the mission, each in its own way. The PID enters a violent but bounded lateral oscillation: the heading doublet triggers it, the heading error grows to $\sim 2100^\circ$ (the aircraft flies repeated full circles), and the altitude sinks by hundreds of meters (Table 8: RMS heading error of 819° , altitude error of 554 m). The gain-scheduled LQRy fails harder: the run diverges outright before 60 s, at the first heading-doublet reversal — its high-bandwidth lateral gains amplify the inverted-sign lateral-directional dynamics instead of containing them: the heading error grows geometrically across the reversal (0.6° at 51 s, 17.7° by 55 s) until the sideslip leaves the physical domain and the simulation terminates. The contrast within the LQ family itself is the most instructive result of the scenario: the earlier, softer fixed-gain LQRy iteration — the build deployed on the bench of Section VI — completes this same mission with near-nominal error (degradation of 0.89–0.99, Table 8 and Fig. 7), so the same structure spans the full range from unconditional survival to outright divergence depending only on its tuning. Robustness to severe mismatch is thus a

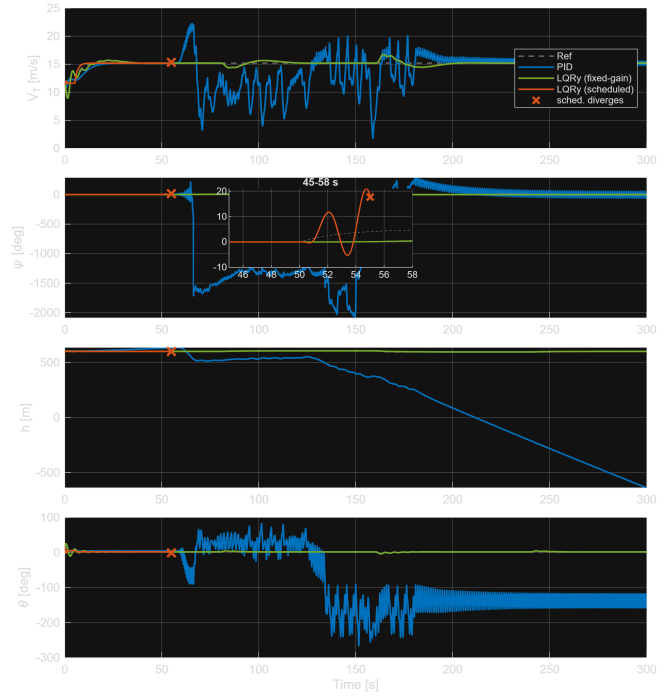


FIGURE 7. Scenario A (severe mismatch): combined mission on the perturbed plant, nominal-design controllers. The fixed-gain LQRy (bench build) completes the mission with near-nominal error; the gain-scheduled LQRy diverges at the first heading reversal ($\times$ marker; inset: 45–58 s).

property of the tuning, not of the LQ structure per se — a point taken up in Section V-E.

B. ISOLATING THE ORIGIN: LONGITUDINAL MISMATCH ONLY

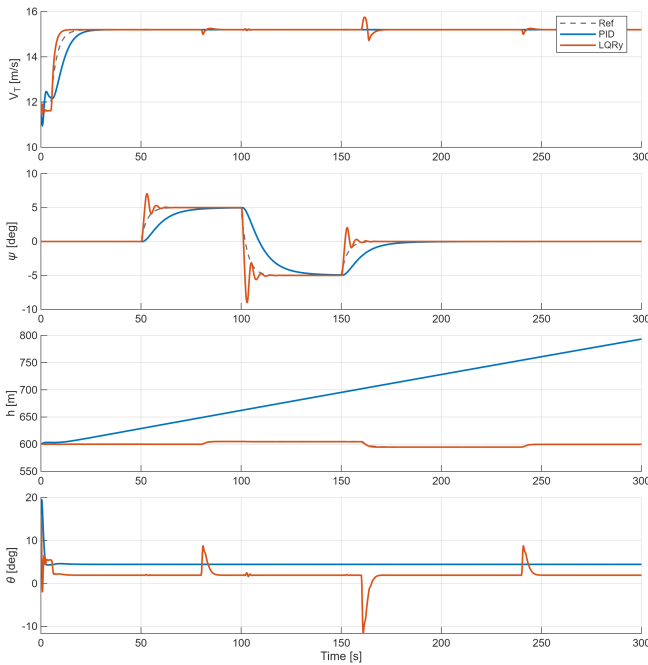
Scenario B replaces only the longitudinal derivatives by the mismatched set, keeping the nominal lateral-directional model. Both laws now complete the mission, which identifies the lateral-directional half of the mismatch — with its sign inversions — as what defeats them both in Scenario A. The LQRy stays near nominal in every channel (degradation of 1.03–1.38). The PID maintains heading and airspeed at nominal levels (degradation of 1.01 and 1.14), but the longitudinal mismatch alone is not benign for it: the altitude channel degrades by a factor of 69 (RMS 113.8 m), a steady drift of the altitude hold. The mechanism is specific: the θ^{ref} clamp of $\pm 10^\circ$ (Section III-B), sized for the control authority of the nominal model, leaves the altitude loop without enough authority on the perturbed plant, and the loop drifts without ever diverging.

C. PLAUSIBLE IDENTIFICATION UNCERTAINTY

Scenario C multiplies all 23 stability and control derivatives by 1.10 — a uniform +10% multiplicative perturbation with preserved signs, representative of a realistic identification error, in the spirit of the perturbed-derivative robustness campaigns of the fixed-wing UAV literature [61]. Both controllers are insensitive to it: every degradation factor lies between 0.82 and 1.03, and the responses are visually indistinguish-

TABLE 8. Robustness campaign: RMS tracking errors ($t > 12$ s) and degradation relative to the nominal scenario (RMS scenario / RMS nominal). Gains and trim always from the nominal design.

Scenario	Ctrl.	RMS error			Degradation ($\times$ nominal)		
		V_T [m/s]	ψ [°]	h [m]	V_T	ψ	h
Nominal	PID	0.071	1.27	1.65	—	—	—
	LQRy	0.040	0.71	0.93	—	—	—
(A) Severe mismatch	PID	3.14	818.7	554.0	44.3	644	335
	LQRy	diverges ^a					
	LQRy (fixed-gain) ^b	0.353	2.61	1.77	0.93	0.89	0.99
(B) Longitudinal only	PID	0.081	1.28	113.8	1.14	1.01	68.9
	LQRy	0.055	0.78	0.96	1.38	1.09	1.03
(C) All derivatives $\times 1.10$	PID	0.073	1.28	1.65	1.03	1.01	1.00
	LQRy	0.032	0.72	0.93	0.82	1.01	1.00
(D) Wind and gusts	PID	0.104	2.24	2.55	1.46	1.76	1.54
	LQRy	0.042	0.75	5.12	1.05	1.06	5.48

^aThe run terminates at ≈ 55 – 60 s, at the first heading-doublet reversal.^bThe earlier fixed-gain bench build (Section VI); degradation relative to its own nominal run.**FIGURE 8.** Scenario B (longitudinal mismatch only).

able from the nominal runs. Within plausible identification uncertainty, both designs are robust; the contrast between them only appears under the severe, sign-changing mismatch of Scenarios A–B.

D. WIND AND GUSTS

Small, low-speed fixed-wing UAVs are disproportionately sensitive to atmospheric disturbance — gust velocities are a large fraction of their airspeed [62] — so no comparison of flight controllers for this class is complete without a wind scenario. Scenario D perturbs the nominal plant through the wind inputs of (3): a 4 m/s tailwind ramped in over 20–25 s, two lateral $1 - \cos$ gusts of ± 3 m/s (55–65 s and 105–115 s,

timed over the heading doublet), and two vertical $1 - \cos$ gusts of ± 2 m/s (85–95 s and 165–175 s, timed over the altitude doublet). The $1 - \cos$ profile is the classical discrete-gust idealization [63], retained to this day in the gust conditions of certification regulations [64]; deterministic discrete gusts (rather than stochastic Dryden or von Kármán turbulence [40], [65]) keep the scenario exactly reproducible run to run, consistent with the deterministic-bench philosophy of Section VI. Both controllers remain stable and complete the mission with moderate peak errors. The LQRy shows the largest single degradation of the scenario — altitude RMS $\times 5.48$ (5.12 m) — while the PID degrades most in heading under the lateral gusts (RMS $\times 1.76$, peak error 12.9°). In absolute terms, however, the LQRy still tracks heading more tightly under the gusts (RMS 0.75° versus 2.24°) but pays in altitude (RMS 5.12 m versus 2.55 m). Neither controller approaches loss of control.

E. DISCUSSION

The study still exposes complementary weaknesses, but re-distributed with respect to the earlier fixed-gain iteration of this comparison. The severity of the model mismatch now defeats both laws: the cascade PID because a sign change in the lateral derivatives invalidates its fixed loop shapes, and the gain-scheduled LQRy because its high-bandwidth lateral gains turn the same sign change into outright divergence. That an earlier, softer fixed-gain LQRy iteration tolerated the same scenario is consistent with the gain and phase margins classically guaranteed for LQ state feedback [12] being lost under output feedback in general [13]: whatever mismatch tolerance an LQ output-feedback design exhibits is an empirical property of the specific tuning rather than a theorem, and aggressive tuning spends it. The discriminators between the two laws lie elsewhere: under the longitudinal-only mismatch (B) the PID silently starves an outer loop through a clamp sized for the nominal plant while the LQRy stays near nominal, and under wind (D) the LQRy con-

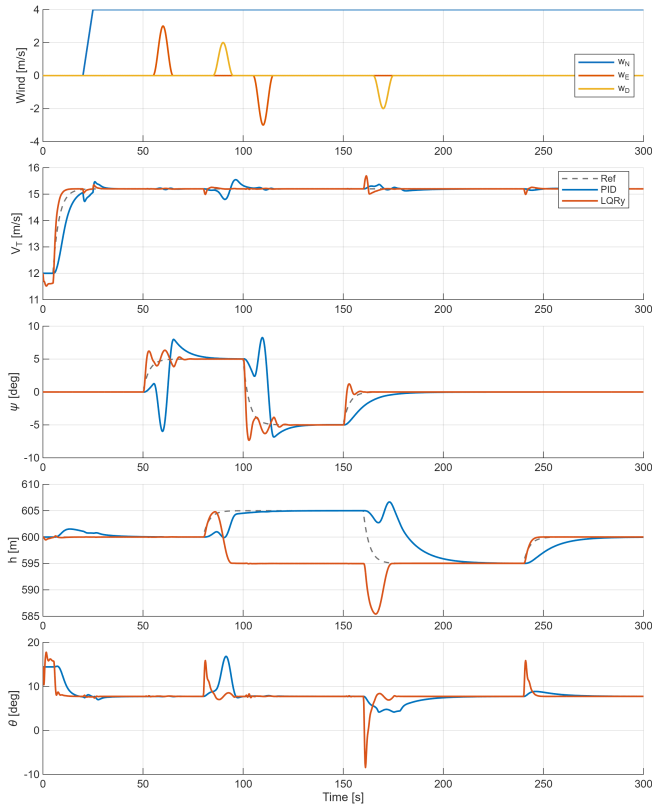


FIGURE 9. Scenario D (wind and gusts): wind profile and closed-loop responses.

cedes altitude where the PID holds firm; within plausible uncertainty (C) both are robust. Robust-control syntheses that shape this trade-off explicitly have been pursued for small fixed-wing UAVs in our own research context [66]; here the point is narrower: between the two standard laws compared, neither dominates under uncertainty, and which weakness matters more depends on whether the dominant threat is model error or atmospheric disturbance.

VI. HARDWARE-IN-THE-LOOP DEVELOPMENT

HIL simulation is the established bridge between simulation and flight for UAV autopilots [14], [15], and the bench lineage of our group runs from the Pegasus real-time HIL architecture [22] through X-Plane-based attitude HIL [23] to the fixed-wing autopilot bench of Santos [24]. This section describes the present incarnation: a lockstep serial bench in which the embedded controller is the auto-generated production code.

A. CONTROLLER DISCRETIZATION

Both controllers are discretized at $T_s = 10$ ms directly in Simulink (automatic conversion of the compensators to their discrete-time forms; the washout filter via Tustin transformation), producing single-rate discrete models suitable for code generation. The discretized PID was verified against the continuous design on the nonlinear model, with an RMS deviation below 0.1% of each command amplitude — dis-

cretization is not a significant error source at this sample rate, as quantified in Section VI-E.

A practical note on output saturation: applying the automatic conversion to the complete control loop with the output saturation blocks included did not yield a working discrete model. The procedure adopted was to discretize the control law alone and to re-insert the saturations after discretization. In the deployment reported here the saturation limits (throttle in $[0, 1]$, surfaces at $\pm 25^\circ$) are enforced in the embedded C code, mirroring the host-side limits exactly.

B. HIL ARCHITECTURE

Fig. 10 shows the architecture. The discrete controller is translated to C by Simulink Embedded Coder (`ert.tlc`, ARM Cortex-M, double precision) — the automatic-code-generation workflow of model-based aerospace software practice [18], [19] — and embedded in an STM32 microcontroller running FreeRTOS — a class of target for which HIL validation remains an active topic [17]. The host PC runs only the nonlinear plant, the reference generation, and the serial interface. Host and target exchange one frame per controller period over a 460 800-baud serial link: 16 single-precision floats to the target (states, filtered references, and mode switches) and 4 floats back (throttle, elevator, aileron, rudder), delimited by a fixed header (`0xDE 0xAD`) and terminator.

The link operates in lockstep, the mechanism by which simulator and autopilot advance strictly by exchanged messages to make runs deterministic and reproducible — standard practice in simulation stacks such as PX4 [67]: the plant blocks at each 10 ms simulation step until the controller reply arrives, so simulated time advances only upon a completed exchange. Every HIL sample therefore corresponds one-to-one to a SIL sample — the comparison is free of timing jitter by construction.

Two aspects of the campaign design are worth stating explicitly, as they are what make the SIL and HIL results directly comparable:

- Single source of references. One script generates both the SIL and the HIL runs of every flight case, so the two campaigns cannot drift apart in their reference definitions.
- Mode selection over the link. The three flight-mode switches travel in the frame rather than being compiled into the firmware, so a single binary serves all flight cases of the PID deployment and the target is never reprogrammed mid-campaign (the LQRy deployment deviates from this, as noted in Section VI-F).

C. LINK AND CHAIN VALIDATION

Before any campaign, each element of the chain was validated independently: (i) serial echo at 460 800 baud with zero errors; (ii) a frame-level test in which the embedded controller response to 200 known input frames matched the simulated discrete controller to within 4×10^{-8} (the single-

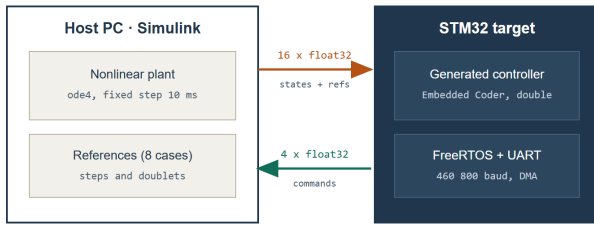


FIGURE 10. HIL loop in lockstep: nonlinear plant on the host, generated controller code on the STM32, binary serial frames at each 10 ms step.

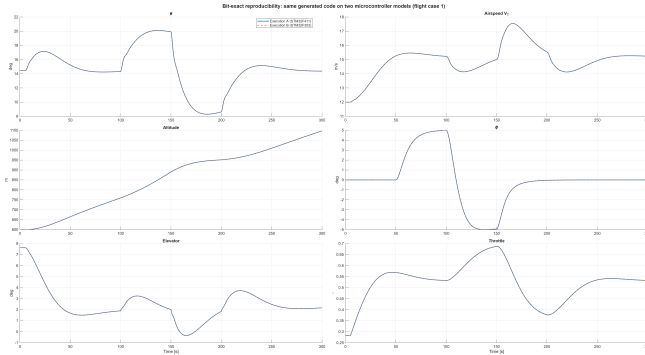


FIGURE 11. Bit-exact reproducibility: the same generated controller code executed on two different microcontroller models over a 300-second flight case. Maximum difference across all channels is identically zero.

precision quantization of the bus); and (iii) the discretization check of Section VI-A.

D. DETERMINISM OF THE LOCKSTEP LOOP

A lockstep HIL loop built from double-precision arithmetic on both sides, with single-precision quantization only on the wire, has no source of run-to-run variability: there is no timing race, no dropped-sample interpolation, and no floating-point reduction whose order can change. The loop should therefore be bit-deterministic.

We verified this property directly. The generated LQRy controller was compiled and executed on a microcontroller model different from the one used in an earlier execution of the same code (STM32F303 versus STM32F411, different core configurations and clock trees), driving the same host model. Over a full 300-second flight case, the two executions produced bit-identical outputs on all recorded channels (maximum difference identically zero, Fig. 11); repeating additional flight cases of the campaign reproduced the same result.

This is a useful property for a validation bench, and not merely a curiosity. Because the loop contributes no noise of its own, any deviation observed between two runs is attributable to a deliberate change in the setup, which makes the bench a measuring instrument rather than a source of scatter. It also means that a campaign can be reproduced exactly by a third party, on different hardware, from the archived code and reference definitions — a stronger form of reproducibility than the usual statistical agreement. The

TABLE 9. Sensitivity of SIL–HIL agreement: each factor injected individually into the validated bench (RMS vs. the clean SIL run, flight case 1).

Factor	θ [°]	h [m]
Baseline (unmodified bench)	0.030	0.09
S1: start-up desynchronization (7 s)	0.014	0.07
S2: airspeed reference mismatch (3 m/s)	0.447	258
S3: controller discretization	0.003	0.001
S4: 1% frame loss	0.032	1.30

sensitivity study of the next subsection relies on this: each experiment changes exactly one factor, and the resulting difference is entirely due to that factor.

This determinism also settles a practical concern of embedded deployment: block execution order in a graphical model can, in principle, alter the generated real-time code with respect to the simulation. Here, with both controllers auto-generated from single-rate discrete models, the cross-MCU bit-exactness rules out target-side nondeterminism, and the quantization-floor SIL–HIL residual of the synchronized PID deployment (Section VI-F) bounds any execution-order effect at the level of the bus quantization itself.

E. SENSITIVITY OF SIL–HIL AGREEMENT

Deploying a control law on an embedded target involves a number of implementation choices whose effect on SIL–HIL agreement is rarely quantified. Exploiting the determinism established above, we injected each factor individually into the validated PID bench and measured the resulting discrepancy against the clean SIL run (Table 9). All experiments use flight case 1; the baseline row is the unmodified bench.

The ranking is informative, and in part counter-intuitive:

1) Reference consistency dominates (S2)

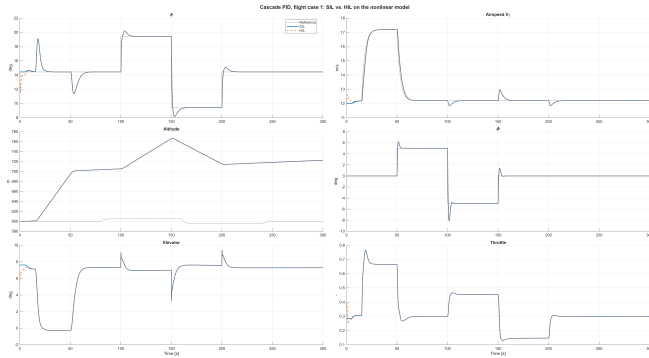
An airspeed reference differing by 3 m/s between the SIL and HIL campaigns — a plausible outcome whenever the two campaigns are configured from separate scripts — produces a 258 m RMS altitude discrepancy. This is not a fidelity loss of the HIL chain at all: it is a comparison between two different flight conditions. It is, by two orders of magnitude, the largest effect measured, which motivates the single-source-of-references design of Section VI-B.

2) Start-up desynchronization leaves a persistent residual (S1)

Embedded controllers commonly hold their integrators in reset until the communication link engages, so the target starts from the trim command while the simulated controller starts from its initial states and produces an initial transient. The two trajectories then differ over the first seconds and, in modes where an outer loop is open, that difference is integrated and never recovered. Synchronizing the two start-ups — as done in the PID deployment — removes this term; leaving them unsynchronized leaves a residual that no amount of link quality can remove.

TABLE 10. PID: RMS SIL–HIL discrepancy per flight case ($t > 12$ s).

Case	θ [°]	V_T [m/s]	h [m]	ϕ [°]
1	0.0028	0.0009	0.089	$\sim 10^{-6}$
2	0.0007	0.0002	0.004	$\sim 10^{-6}$
3	0.0035	0.0011	2.75	$\sim 10^{-6}$
4	0.0007	0.0002	0.004	$\sim 10^{-7}$
5	0.0109	0.0059	0.402	$\sim 10^{-6}$
6–8	$< 10^{-3}$	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-6}$

**FIGURE 12.** PID, flight case 1: SIL vs. HIL on the nonlinear model.

3) Frame loss is benign at low rates (S4)

Discarding 1% of the controller replies at random — each discarded reply is replaced by holding the previous command for that step, so the lockstep timing is preserved — degrades altitude agreement to 1.30 m but leaves attitude essentially unaffected, since the actuator rate limits and the airframe dynamics filter isolated command glitches. This is reassuring for links less reliable than the one used here.

4) Discretization is negligible (S3)

The usual first suspect — the continuous-to-discrete conversion of the control law — accounts for 0.003° in attitude and 10^{-3} m in altitude at $T_s = 10$ ms, three orders of magnitude below the reference-consistency term.

F. EMBEDDED CAMPAIGN RESULTS

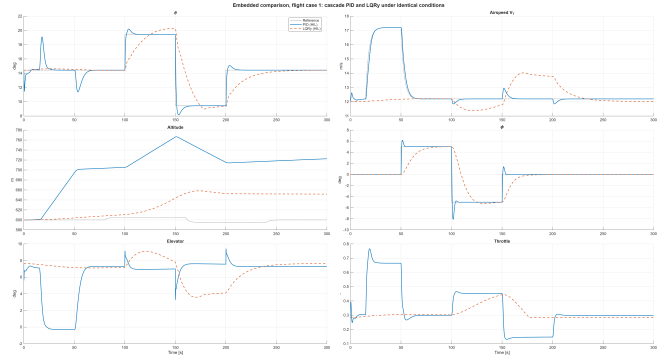
Eight flight cases (all combinations of the three mode switches, with doublets on attitude/heading and steps on airspeed/altitude, 300 s each) were executed in SIL and HIL for both controllers, from a single script per controller.

1) PID

Table 10 reports the RMS SIL–HIL discrepancy ($t > 12$ s). The residual reaches the quantization floor: pitch attitude within 0.011° in the worst case, altitude within 2.75 m only where the altitude loop is open (integrated drift), and essentially zero elsewhere. Fig. 12 illustrates case 1: the SIL and HIL traces are indistinguishable at plot scale.

2) LQRy

The embedded LQRy is the earlier fixed-gain delivery — a softer gain set than the gain-scheduled design of Sec-

**FIGURE 13.** Embedded comparison, flight case 1: PID and LQRy under identical conditions on the same bench.

tion III-C, registered in the repository [30] — so the bench results validate the deployment pipeline for that build; re-deploying the gain-scheduled law on the bench is future work (Section VII). The LQRy campaign was executed under the same protocol, with per-case firmware builds and scripted programming of the target. Its SIL–HIL residual is larger — from 0.23° to 3.57° in pitch depending on the flight case — and is explained by the start-up term quantified in S1: this implementation holds the embedded integrators in reset until the link engages, so the embedded and simulated runs begin from different states. The effect is deterministic and confined to the transient; the doublet responses themselves track closely.

G. CONTROLLER COMPARISON ON THE BENCH

With both controllers on the same bench, the same plant, the same references, and the same metric window, the dominant free variable is the control law — subject to the harness asymmetries disclosed in Section IV-B, which the equalization ablation showed not to drive the conclusions. Fig. 13 shows flight case 1: on the bench, the PID tracks airspeed and pitch more tightly, while the LQRy regulates altitude better when the airspeed loop is active — recalling that the embedded LQRy is the earlier fixed-gain build, so these bench observations characterize that deployment rather than the gain-scheduled law of Section IV-E.

VII. CONCLUSION

This article covered the full path from controller design to embedded validation for the fixed-wing flight mode of a hybrid VTOL aircraft. A cascade PID and an output-feedback LQR were designed on the linearized model, verified on the nonlinear model, stressed under model mismatch and wind, and deployed to an STM32 microcontroller through automatic code generation, so that the artifact exercised on the bench is the production code itself.

Four results stand out. First, on the nominal plant the gain-scheduled LQRy is the faster law — guidance captures in 6–13 s and RMS tracking errors roughly $2\times$ smaller — but violates the heading-overshoot requirement (24.5%), while the requirements-retuned PID meets every requirement row

at lower bandwidth. Second, the robustness study shows that a severe, sign-changing lateral-directional model mismatch defeats both compared laws — the aggressive gain-scheduled LQRy diverging outright, while the earlier, softer fixed-gain iteration of the very same structure completes the mission with near-nominal error — which exposes robustness to severe mismatch as a property of the tuning rather than of the LQ structure; the remaining scenarios still reveal complementary weaknesses: under the longitudinal-only mismatch the LQRy stays nominal while the PID's clamped altitude loop drifts, under wind and gusts the LQRy concedes altitude (5.1 m RMS) where the PID stays firm, and both tolerate a plausible +10% identification uncertainty. Which weakness matters more depends on whether the dominant threat is model error or atmospheric disturbance. Third, the lockstep HIL architecture is practically transparent when the campaign is scripted from a single source of references and the controller start-up is synchronized: for the synchronized PID deployment, the SIL–HIL residual reaches the quantization floor of the communication bus (0.011° in pitch attitude in the worst flight case), and the loop is bit-deterministic — the same generated code executed on two different microcontroller models reproduces full 300-second flight cases bit for bit — which makes the bench a measuring instrument and allows a campaign to be reproduced exactly on different hardware. Fourth, exploiting that property, a sensitivity study ranked the implementation choices that actually matter for SIL–HIL agreement: reference consistency between campaigns dominates by two orders of magnitude, start-up desynchronization leaves a persistent residual that link quality cannot remove, low-rate frame loss is benign, and controller discretization — the usual first suspect — is negligible at a 10 ms sample rate.

Future work follows directly from the limitations identified here: a feedforward decoupling term for the roll-to-pitch coupling quantified in Section III-D, evaluated on both control laws; an extended robustness campaign with stochastic Dryden/von Kármán turbulence [40], [65] in addition to the discrete gusts used here, and with robust syntheses designed for this trade-off [66] as a third contender; equalization of the two design harnesses (actuator models, reference shaping, and protection logic) so that future comparisons are free of these asymmetries; a requirements set rederived specifically for unmanned aircraft in the sense of the scaled flying-qualities literature [46], [59]; re-deployment of the gain-scheduled LQRy on the bench, so that the embedded comparison covers the same build as the SIL comparison; and flight testing of the embedded controllers validated on the bench, closing the simulation-to-flight loop that the HIL stage approximates [60].

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